

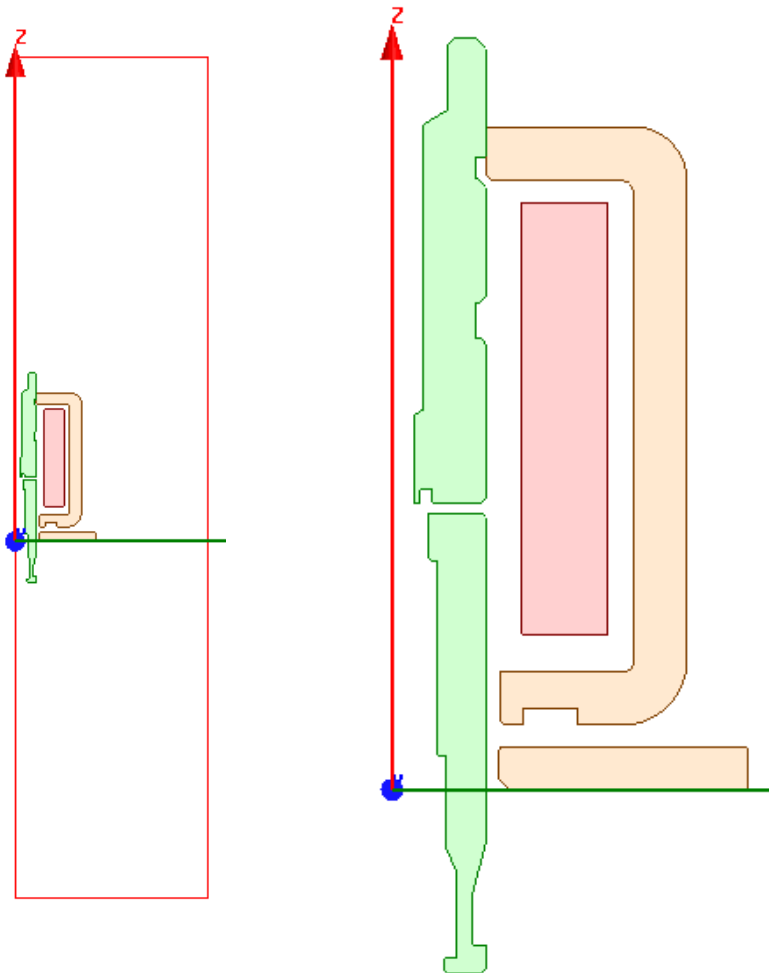
결과 비교 (2D RZ plane vs. 3D)

Think with TSNE
Solve with ANSYS
Network with Customers
Enjoy with Results

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(Taesung S&E)

2D RZ Plane

Model



Material

- Sheets
 - copper
 - + coil
 - SR430
 - + Armature
 - + Core
 - + Top_Plate
 - + Yoke
 - vacuum
 - + Region

Excitation & Results (Force, Matrix)

Excitation

Project Manager

- Solenoid_1210_2014
 - 2D_RZ_MS (Magnetostatic, about Z)
 - Model
 - Boundaries
 - Excitations
 - Current1
 - Parameters
 - Mesh Operations
 - Analysis
 - Optimetrics
 - Results
 - Field Overlays
 - 2D_RZ_TR (Transient, about Z)
 - Definitions

Project

Properties

Name	Value	Unit	Evaluated V...
Name	Current1		
Type	Current		
Current	(24/195)*3700		455,38461538...
Direction	Positive		

Force, L-Matrix

Profile | Convergence | Force | Matrix | Mesh Statistics

Parameter: Amature_F Force Unit: newton

Pass: 3

	F(z)	
Total	8.3904	

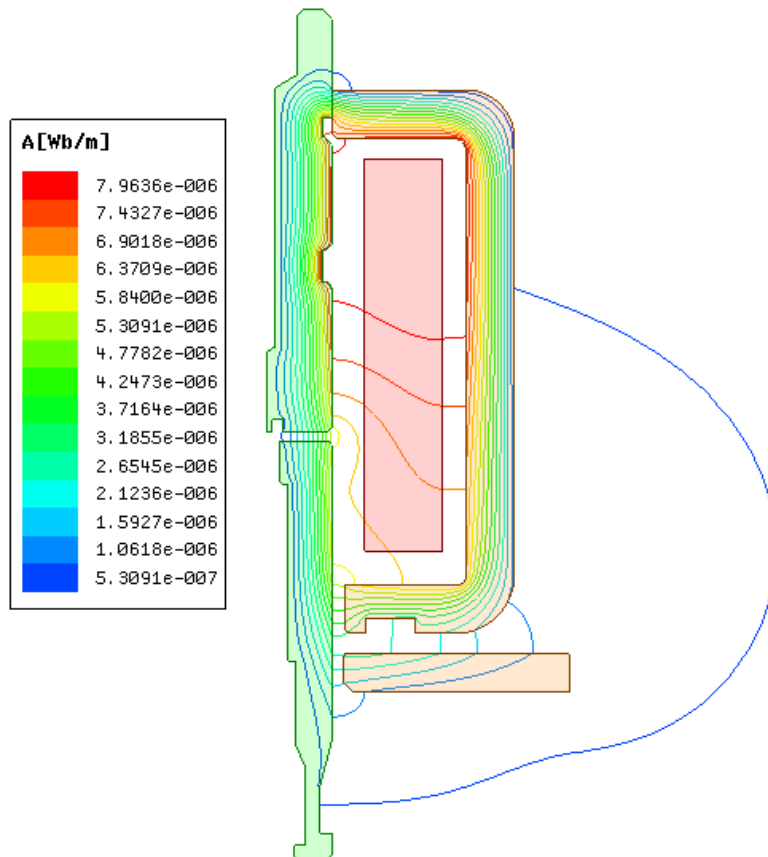
Profile | Convergence | Force | Matrix | Mesh Statistics

Parameter: Matrix1 Type: Inductance

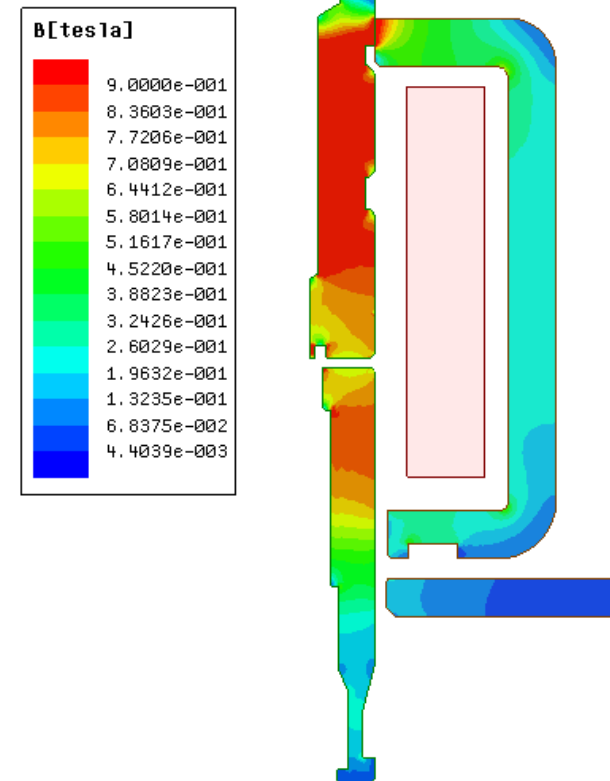
Pass: 3 Inductance Units: mH

	Current1	
Current1	1438.1	

Flux Line



B magnitude

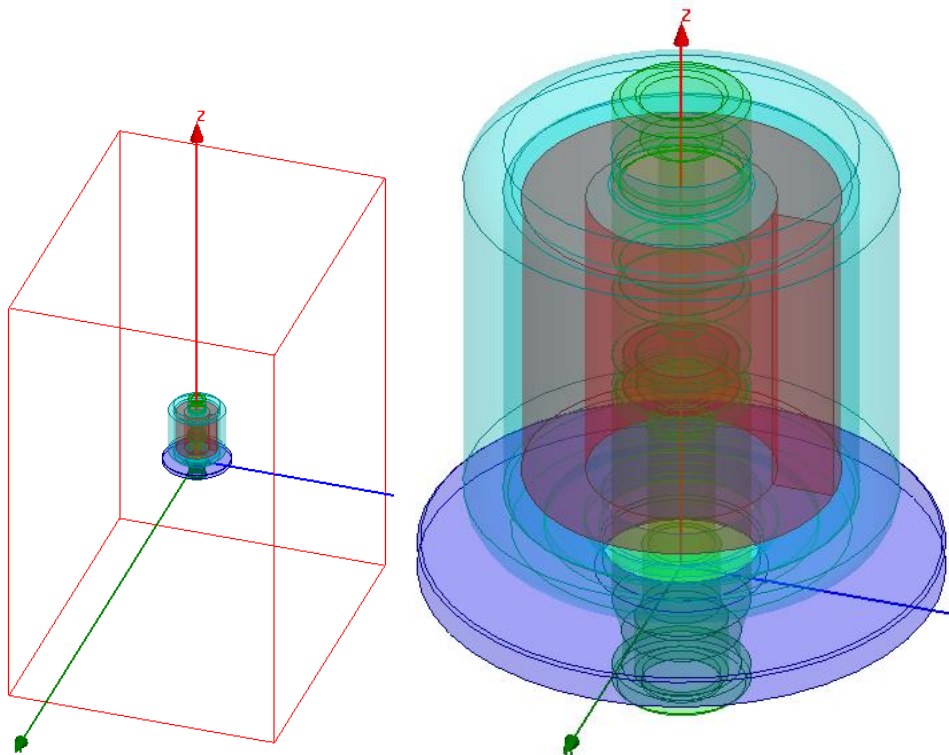


3D model from 2D RZ model

Model & Materials

Model

Material



- Sheets
 - copper
 - + coil
- SR430
 - + Armature
 - + Core
 - + Top_Plate
 - + Yoke
- vacuum
 - + Region

Excitation & Results (Force, Matrix)

Excitation

Project Manager

- Solenoid_1210_2014_add*
 - 2D_RZ_MS (Magnetostatic, about Z)
 - 2D_RZ_TR (Transient, about Z)
 - 3D_Model_from_2D (Magnetostatic)*
 - Model
 - Boundaries
 - Excitations
 - Current1
 - Parameters
 - Mesh Operations
 - Analysis
 - Setup1
 - Optimetrics
 - Results
 - Field Overlays

Project

Properties

Name	Value	Unit	Evaluated V...
Name	Current1		
Type	Current		
Current	(24/195)*3700	A	455,38461538...
IsSolid	Stranded		
Direction	Point into terminal		

Force, L-Matrix

Profile | Convergence | Force | Torque | Matrix | Mesh Statistics

Parameter: Force Unit:

Pass:

	F(x)	F(y)	F(z)	Mag(F)
Total	-7.4948E-005	-0.0015382	8.5593	8.5593

Profile | Convergence | Force | Torque | Matrix | Mesh Statistics

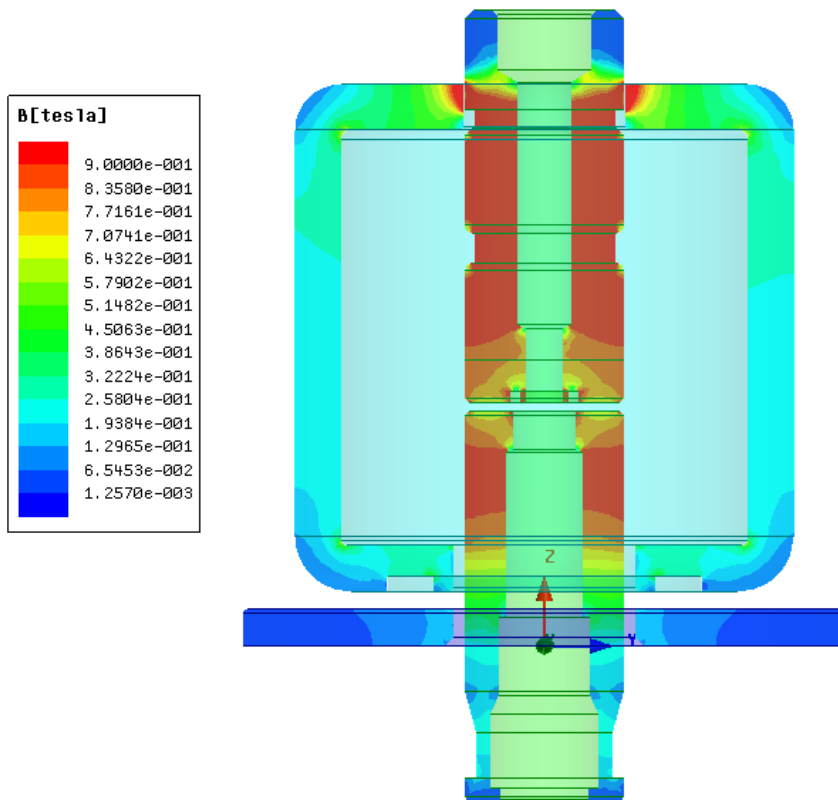
Parameter: Type:

Pass: Inductance Units:

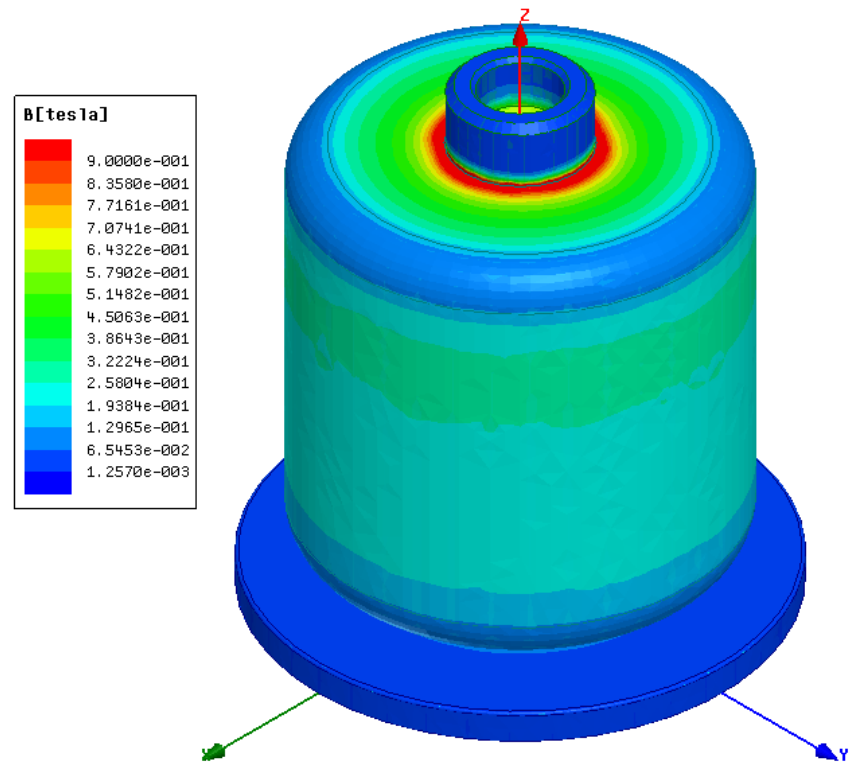
	Current1
Current1	1463.1

Plot – B magnitude

YZ-plane



3D view

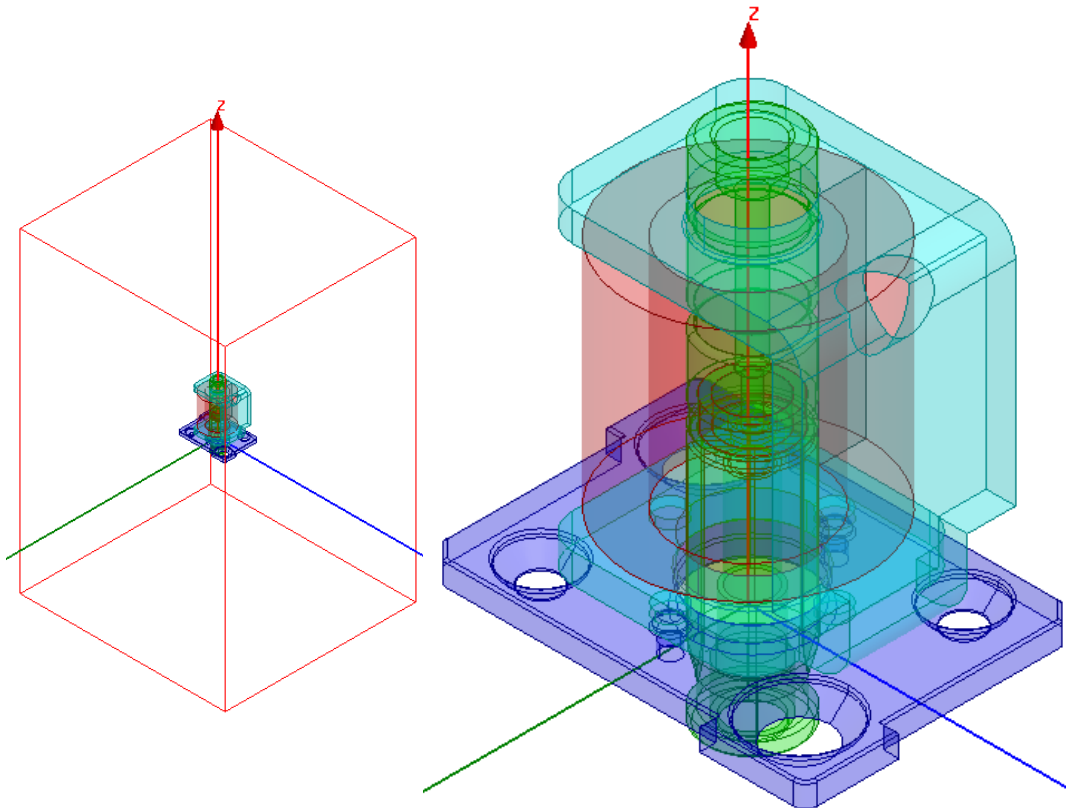


3D Model from TPC

Model & Materials

Model

Material



- Sheets
 - copper
 - + coil
- SR430
 - + Armature
 - + Core
 - + Top_Plate
 - + Yoke
- vacuum
 - + Region

Excitation & Results (Force, Matrix)

Excitation

Project Manager

- Solenoid_1210_2014_add*
 - 2D_RZ_MS (Magnetostatic, about Z)
 - 2D_RZ_TR (Transient, about Z)
 - 3D_Model_from_2D (Magnetostatic)*
 - 3D_Model_from_TPC (Magnetostatic)*
 - Model
 - Boundaries
 - Excitations
 - Current1
 - Parameters
 - Force1
 - Matrix1
 - Mesh Operations
 - Analysis
 - Setup1

Project

Properties

Name	Value	Unit	Evaluated V...
Name	Current1		
Type	Current		
Current	Input_current		455.38461538...
IsSolid	Stranded		
Direction	Point into terminal		

Force, L-Matrix

Profile | Convergence | Force | Torque | Matrix | Mesh Statistics

Parameter: Force1 Force Unit: newton

Pass: 6

	F(x)	F(y)	F(z)	Mag(F)
Total	-0.0015032	0.043226	5.0275	5.0277

Profile | Convergence | Force | Torque | Matrix | Mesh Statistics

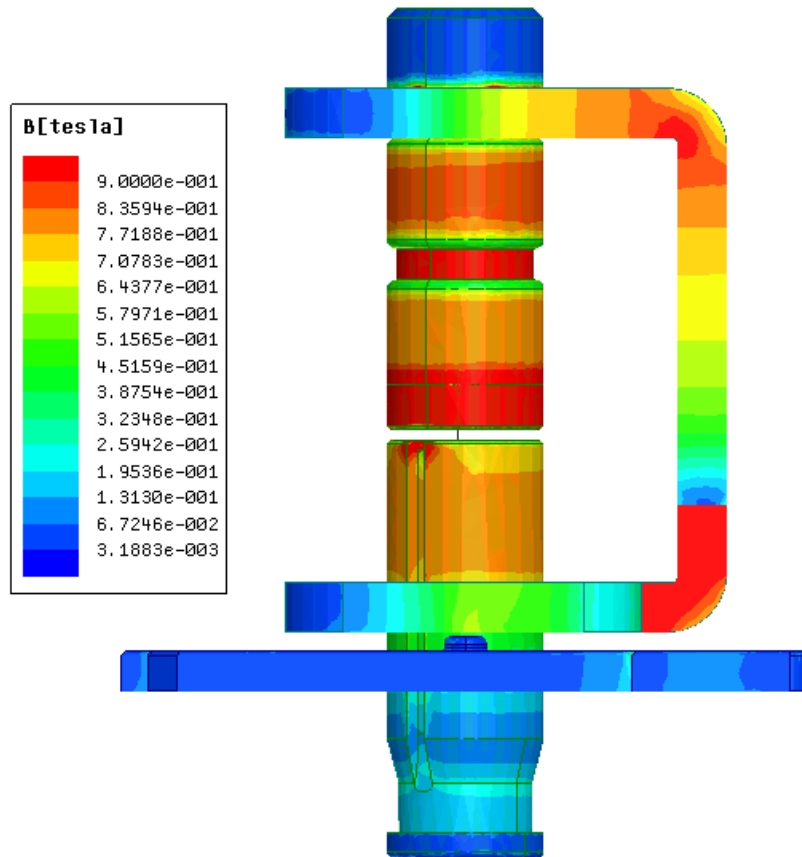
Parameter: Matrix1 Type: Inductance

Pass: 6 Inductance Units: mH

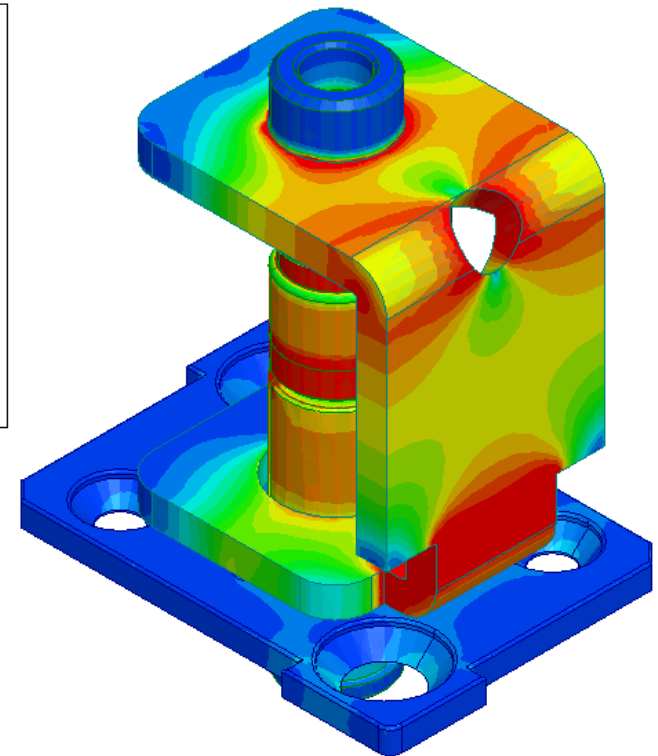
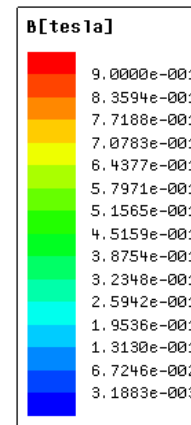
	Current1
Current1	1074.6

Plot – B magnitude

YZ-plane



3D view



Summary

	FORCE (Fz) [N]	Inductance [mH]
2D RZ model	8.3904	1438.1
3D model from 2D RZ	8.5593	1463.1
3D model from TPC	5.0275	1074.6

- 2D RZ Plane 결과와 이 형상으로 3D 를 만든 모델의 결과는 매우 유사함.
- 원 모델의 결과는 이와 매우 상이함을 알 수 있음.
 - ➔ Yoke 가 일부분만 있는 것
 - ➔ Hole 들이 뚫려 있는 것
 - ➔ core 나 armature 부분이 매끄러운 원통이 아닌 형상이기 때문.

감사합니다!